

STRENGTH PERFORMANCE IN YOUTH: TRAINABILITY OF ADOLESCENTS AND CHILDREN IN THE BACK AND FRONT SQUATS

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ABSTRACT

Keiner, M, Sander, A, Wirth, K, Caruso, O, Immesberger, P, and Zawieja, M. Strength performance in youth: trainability of adolescents and children in the back and front squats. *J Strength Cond Res* 27(2): 357–362, 2013—A basic question for many athletic coaches pertains to the maximum attainable strength level for youth athletes. The aim of this investigation was to establish reference values for the strength performance in the front and back squats in youth athletes. The strength performance in front and back squats of 141 elite youth soccer players was tested by a 1 repetition maximum (1RM) and 1RM related to bodyweight (SREL). The subjects aged between 11 and 19 years and were divided into 2 groups and 4 subgroups (A = younger than 19 years, B = younger than 17 years, C = younger than 15 years, and D = younger than 13 years). For approximately 2 years, one group (control group [CG]) only participated in routine soccer training and the other group (strength training group [STG]) participated in an additional strength training program along with the routine soccer training. Additionally, the strength performance in a 5RM in both squat variants of 105 young elite weightlifters (National Weightlifting Organization Baden Württemberg, [BWG]) was examined to show the high level of trainability of children and adolescents and to determine the reference strength values for young athletes. The STG performed in the parallel front squat SREL in the subgroups A 1.7 $\pm$ 0.2, B 1.6 $\pm$ 0.2, C 1.4 $\pm$ 0.2 and D 0.9 $\pm$ 0.3. The STG had significantly ($p < 0.001$) higher strength values in 1RM and SREL than CG. The BWG had higher strength values than STG, but the BWG was not part of the statistical analysis because of the different test protocols (1RM vs. 5RM). Our data show that the SREL in parallel squat for young elite athletes with long-term training

experience should be a minimum of 2.0 for 16- to 19-year-olds, 1.5 for 13- to 15-year-olds, and 0.7 for 11- to 12-year-olds.

KEY WORDS soccer, weightlifting, 1RM, strength performance

INTRODUCTION

Physiologically, the maximum strength and power of a muscle depend on its cross-sectional area and the neuronal activation. As such, achievement of maximum muscle strength is largely because of an increase in these parameters. Because research has documented average-to-strong relationships between strength parameters and sprint and jump performance (2,3,9,10,12,13,21,25,26), improving muscle strength can be particularly crucial to athletes. Several different methods have been used to assess strength in the investigation of its relationship to athletic performance. These include isokinetics, machine squats, and free-weight squats (1,9,12,25). Strong correlations between speed and both absolute and relative strength have been found in the use of free-weight squats (1,12,25). For example, Wisloff et al. (25) demonstrated a strong relationship between free-weight squat and jump performance, and additional investigators have shown a positive relationship between improved strength parameters and jump performance (13). Furthermore, improved muscle strength has been shown to reduce athletic-related injuries (6). In many sports, body weight must be moved quickly during sprints and jumps. Therefore, instead of primarily recording absolute strength values, strength performance in relation to body weight (SREL) should also be evaluated. The literature provides data for SREL in adults. The SREL values for football players are between 1.7 and 1.9 in the parallel back squat (1,12) and between 1.1 and 1.3 in the parallel front squat (10). Wisloff et al. (25) found SREL values of 2.2 for soccer players in the half squat. Reference values for youth could not be found in a review of the literature.

Long-term periodized strength training is recommended for young athletes, but many coaches, parents, and athletes worry that beginning strength training too early can increase the risk of injury. However, fewer injuries have been reported

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in weightlifting compared with other sports despite the use of heavy weights among those athletes (5,8,16,17,18). Although back and knee injuries are commonly discussed, they are not the most common injuries sustained during weightlifting (11,15). Most injuries occur because of a lack of attention, improper handling, and improper supervision (5,15). Furthermore, there are no scientifically supported findings that suggest that long-term strength training leads to damage to the musculoskeletal system or affects growth (4). Instead, as reported by Faigenbaum et al. (4), there is evidence that strength training can have positive effects, such as maximizing bone mineral density during childhood and adolescence. Thus, childhood and adolescence may actually be the ideal time to develop the coordination and skill techniques needed to acquire high maximum strength values. The earlier one begins long-term periodized strength training, the easier and faster it is for that individual to reach high strength values. A basic question then exists regarding the maximum strength level attainable for youth athletes. The aim of this investigation was to establish reference values for the strength performance in the front and back squats in youth athletes and to demonstrate the high level of trainability of youths.

METHODS

Experimental Approach to the Problem

The objectives of this study were to establish reference values for the strength performance of youth athletes and to show the high level of trainability of children and adolescents. To accomplish these objectives, soccer players with and without experience in strength training were measured and compared. Strength performance in relation to body weight and 1 repetition maximum (1RM) in the back and front squats were designated as dependent variables. The presence or absence of strength training experience (STG and CG, respectively) was designated as the independent variable. Furthermore, for an improved rating in soccer players' strength values, reference values for the front and back squats in youth weightlifters were consulted. These values were not included in the statistical analysis.

Subjects

Elite youth soccer players ($n = 141$) and national elite youth weightlifters ($n = 105$; BWG) volunteered for this investigation. The soccer players were recruited from 2 youth training centers affiliated with professional teams in the second and third divisions in Germany, whereas the weightlifters were recruited from a national weightlifting organization (Baden-Württemberg). Each subject was informed of the experimental risks involved with the research. All subjects provided written informed consent to participate. Informed consent was also obtained from the subjects' parents, where subjects were younger than 18 years. The research design was approved by the institutional review board of the Institute of Sport Science, Johann Wolfgang Goethe - University for the use of human subjects. The subjects were between 11 and 19 years old. The

youth weightlifters and the youth soccer players were divided into cohorts D, C, B, and A. These cohorts included 11–12, 13–14, 15–16, and 17–19 years of age, respectively. The soccer players were then divided into 2 subgroups: those who had participated in additional strength training programs for approximately 2 years (STG) and those who had only performed their routine soccer training (CG). A further control group of youth who were not engaged in competitive sports was not included because this was too difficult to organize.

Procedures

Training Protocol. The subjects in the STG group had strength training experience of about 2 years. Strength training in the STG group was completed twice a week, in addition to the 4 soccer training sessions. Subjects in the A, B, and C cohorts varied their exercises between the parallel front and back squats during the week. Additionally, the players of these cohorts performed bench presses, deadlifts, neck presses, and exercises for the trunk muscles, as well as the standing row. The strength training for the squat was periodized, and the subjects performed technique training for squats during the first 4 weeks. During the following 8 weeks, they performed a hypertrophy training consisting of 5 sets of 10 repetitions, with a rest of 3 minutes between sets. The third training block consisted of 5 sets of 6 repetitions and a rest of 3 minutes between sets, followed by an additional training block marked by 5 sets of 4 repetitions and a rest of 5 minutes between sets. Faigenbaum et al. (4) showed that this training protocol is appropriate for youth. The amount of lifted weight in the training sessions was determined by the ability of each individual person, and the subjects had to lift their maximum weight while using the correct technique for the given repetitions (4, 6, or 10RM). The amount of weight was increased at the next training session if the subject could lift the given weight with the proper technique in the respective squat variant. The training for the trunk muscles and the upper extremities always consisted of 3–5 sets of 10 repetitions and a rest of 3 minutes between sets. The subjects always performed a rotation of 3 of the previously mentioned exercises for the upper body and 1 exercise for the trunk muscles, in addition to the squat exercise, during every training session. The described periodization was repeated twice during the season. Cohort D participated in technique-oriented training that consisted of 3 sets of squats, 10 repetitions, and 3 minutes of rest between sets once a week in the weight room throughout the year. The parallel front and back squat exercises varied during the training. The weight was increased very carefully, and correct technique was the focus of attention. Additionally, the players performed bench presses, deadlifts, neck presses, and exercises for the trunk muscles, as well as the standing row. The second additional strength training was performed once a week during the regular soccer training. This training consisted of 3 sets of lunges with 10 repetitions and 3 minutes of rest between sets and 2 exercises for the trunk.

The CG only participated in 4 regular soccer training sessions per week, and none of the subjects participated in an additional strength training routine. The training program of the BWG followed a half-year periodization with 5–8 training sessions per week. The half-year periodization starts with high volume and moderate intensity of lifts (up to 500 repetitions per week) and ends with a low volume and high intensity of lifts (up to 200 lifts per week) of different exercises. Within the half-year periodization, the weightlifters followed a 3-week lasting cycle. Within this cycle, the intensity tends to be the same, but the routine follows 2 weeks of high volume and 1 week of moderate volume. The pool of exercises includes the clean, clean and jerk, and snatch in addition to other supplemental exercises, such as the clean pull, snatch pull, squat, and front squat.

Testing Protocol. The warm-up was standardized for the soccer players and weightlifters and consisted of 2 sets of squats with 6–8 repetitions with a submaximal weight. The strength tests were evaluated in a maximum of 5 trials. The maximum strength of the front squat was measured first, followed by the back squat. The soccer players participated in technique training for the squat variants twice a week for 2 weeks before they were tested. No subjects participated in a fatiguing training session for a minimum of 2 days before testing. None of the participants reported any injury at the time of testing. Anthropometric and performance measurements were collected by the same researchers at the same time on testing day, and all participants were asked to wear the same clothing and footwear. All participants were asked to eat and drink a sufficient amount until 1 hour before testing.

Some limitations to our study design exist. For example, soccer players were tested in a 1RM, and the weightlifters were tested in a 5RM. Along with this difference in strength

TABLE 1. Body weight and height of the soccer players.

Group	Subgroup (n)	Weight (kg)	Height (cm)
STG (n = 77)	A (13)	72.8 ± 7.7	176.6 ± 6.8
	B (30)	70.1 ± 9.1	177.4 ± 9.1
	C (18)	54.5 ± 10.0	166.5 ± 5.9
	D (16)	45.4 ± 7.8	157.3 ± 9.6
CG (n = 64)	A (11)	76.0 ± 6.5	178.8 ± 6.4
	B (21)	70.0 ± 9.0	178.6 ± 7.2
	C (19)	56.8 ± 9.1	169.6 ± 7.2
	D (13)	42.0 ± 2.9	154.2 ± 4.7

tests between groups (5RM vs. 1RM), there could be different variations of squats depending on which technique was used. The squat could be performed as the quarter squat, the half squat, the parallel squat, or the depth squat. The difference is in the range of motion. The turning point in the quarter squat is at a knee angle of 120°, in the half squat at 90°, in the parallel squat at 60°–70°, and in the depth squat at 45°–60°, depending on thigh and hamstring size. The depth squat requires less weight to generate an adequate stress stimulus for the lower extremities compared with the other variations of the squat. The weightlifters performed a depth squat during their strength tests, but the soccer players were not able to perform a depth squat because of their inflexibility. Thus, the soccer players performed a parallel squat instead. To compare the values of the soccer players and the weightlifters, the 4RM from the last training session of

TABLE 2. Strength values of the soccer players in the front squat.*

Group	Variant subgroup (n)	1RM (kg)	1RM SREL	4RM (kg)	4RM SREL
STG (n = 77)	A (13)	106.9 ± 9.9†	1.5 ± 0.2†	81.7 ± 7.5	1.1 ± 0.1
	B (30)	97.8 ± 13.3†	1.4 ± 0.2†	79.5 ± 8.1	1.2 ± 0.2
	C (18)	65.3 ± 17.1†	1.2 ± 0.2†	49.7 ± 12.9	1.0 ± 0.2
	D (16)	30.0 ± 5.5†	0.7 ± 0.2†		
CG (n = 64)	A (11)	62.7 ± 9.6†	0.8 ± 0.1†		
	B (21)	51.2 ± 17.1†	0.7 ± 0.2†		
	C (19)	29.2 ± 7.3†	0.5 ± 0.1†		
	D (13)	15.4 ± 5.6†	0.4 ± 0.1†		

*1RM = 1 repetition maximum; SREL = 1RM/body weight.

†Significant ($p < 0.001$) between the groups.

TABLE 3. Strength values of the soccer players in the back squat.*

Group	Variant subgroup (n)	1RM (kg)	1RM SREL	4RM (kg)	4RM SREL
STG (n = 77)	A (13)	121.5 ± 11.1†	1.7 ± 0.2†	98.8 ± 11.1	1.3 ± 0.2
	B (30)	113.0 ± 15.2†	1.6 ± 0.2†	93.7 ± 12.5	1.4 ± 0.2
	C (18)	75.0 ± 18.7†	1.4 ± 0.2†	59.1 ± 16.3	1.1 ± 0.2
	D (16)	41.3 ± 10.4†	0.9 ± 0.3†		
CG (n = 64)	A (11)	85.9 ± 9.4†	1.1 ± 0.1†		
	B (21)	62.1 ± 12.1†	0.9 ± 0.2†		
	C (19)	37.2 ± 6.9†	0.7 ± 0.1†		
	D (13)	17.7 ± 6.3†	0.4 ± 0.1†		

*1RM = 1 repetition maximum; SREL = 1RM/body weight.

†Significant ($p < 0.001$) between the groups.

the STG was examined. The 4RM of the soccer players was not included in the statistical analysis.

The maximum strength of the lower extremities was tested for the soccer players by a 1RM in the parallel front and back squats. The test-retest correlation (r) was 0.97 ($p = 0.001$) for the front squat and 0.99 ($p = 0.001$) for the back squat. The 1RM method is recommended for testing the maximum strength of soccer players and youth (4,22). The testing protocol for the

soccer players occurred during June 2011, 2 weeks after the last match of the season.

The weightlifters were tested with a 5RM depth squat for the front and back squats, but these data were not included in the statistical analysis. These data values were only used as reference values for the strength performance of the soccer players. The 5RM for squats is a commonly used test procedure for determining a reference value in youth weightlifting and is tested minimum 2 times per year (at the end of half-year periodization). The values for each cohort originated from the best athletes (minimum of 5) among the BWG in the last 5 years (2006–11).

Statistical Analyses

The data were analyzed using SPSS 19.0. For the soccer players ($n = 141$), the Kolmogorov-Smirnoff test for normal distribution confirmed the normality of that group's data. A t -test for independent samples was used to determine the differences in each cohort between the STG and CG. A significance level of $p \leq 0.05$ was used. The Levene Test of Homogeneity could be used because of the different sample sizes. Subsequently,

TABLE 4. Absolute and SREL strength performances (5RM) of the young weightlifters ($n = 105$) in the back and front squats.*

Group	Body weight category (kg)	Front (kg)	Back (kg)	Front SREL	Back SREL
A	–62	115	135	1.9	2.2
	–69	124	144	1.8	2.1
	–77	141	155	1.8	2.0
	–85	150	166	1.8	2.0
B	–56	82	89	1.5	1.6
	–62	95	104	1.5	1.7
	–69	110	120	1.6	1.7
	–77	119	136	1.5	1.8
C	–85	134	150	1.6	1.8
	–40	49	60	1.2	1.5
	–45	54	64	1.2	1.4
	–50	60	69	1.2	1.4
D	–56	65	75	1.2	1.3
	–62	70	85	1.1	1.4
	–30	15	20	0.5	0.7
	–35	18	24	0.5	0.7
	–40	25	30	0.6	0.8
	–45	30	40	0.7	0.9
	–50	36	44	0.7	0.9

*5RM = 5 repetition maximum; SREL = 5RM/body weight.

according to Cohen, the effect size of each variable was calculated [$d = (M1 - M2) / \sqrt{(SD1^2 + SD2^2) / 2}$]. In general, effect sizes higher than 0.5 are considered large, effect sizes from 0.3 to 0.5 are considered moderate, effect sizes from 0.1 to 0.3 are considered small, and effect sizes less than 0.1 are trivial.

RESULTS

The anthropometric results are shown in Table 1. There was no difference in body weight and size between the groups in the A, B, C and D cohorts.

Statistical analyses of the age cohorts (A, B, C, and D) in the soccer group showed significantly ($p \leq 0.001$) higher values of the STG compared with the CG in all tested variables (Tables 2 and 3). The effect size for all of the differences in variables was $d > 1.5$. The strength values for the weightlifters are presented in Table 4.

DISCUSSION

To our knowledge, this is the first study to evaluate possible reference values in strength performance in young athletes. The strength values of the STG and the BWG are a starting point for referencing strength performance of young elite athletes of different sports. The data of the BWG shows the high level of trainability.

To determine reference values of 1RM and SREL for young athletes for each age cohort, we also examined values for the youth weightlifters. These weightlifters' values were not included in the statistical analysis and should only be recognized as reference values of strength-trained "specialists." The reference values for strength provided a better rating of the soccer players' results. The youth weightlifters showed higher strength values in every cohort when compared with the soccer players. Furthermore, the values showed that adolescents are able to lift large amounts of weight, which demonstrates a high level of trainability at this age. Between 17 and 19 years of age, the weightlifters reached SREL values of up to 2.2 in a 5RM. This 5RM value exceeded the 1RM value of the soccer players (20). It must be noted that the weightlifters performed a depth squat, whereas the soccer players only performed a parallel squat. Generally, deeper squats correspond to a smaller amount of lifted weight. Therefore, when comparing the values of the soccer players with those of the weightlifters, the achieved values of the soccer players must be put in perspective. When examining the 4RM values, the predominance of the weightlifters compared with the STG became more apparent. In this comparison, the lifted weight among the weightlifters was considerably higher than that of the soccer players. However, the values for the weightlifters should not be used as reference points for soccer players or athletes in other sports. They should instead be used to demonstrate the high level of trainability of adolescents. Based on the evaluated values in this investigation (Tables 2 and 3), an SREL of minimum 2.0–2.5 in the front and back squats for strength-trained youth athletes can be suggested. Long-term

periodization must occur to reach these reference values, and the earlier that the strength training begins, the easier and faster it is to reach these reference points. It is conceivable for soccer players, weightlifters, and athletes from other sports to start strength training at an early age. The youth athletes should begin strength training at 7–8 years of age so that in the best-case scenario, they may have up to more than 10 years of weightlifting experience by the time they are 19 years old. The SREL in squat performance for soccer players, and athletes from other sports with long-term training experience, should be, in the opinion of the authors, a minimum of 2.0 for the A and B cohort, 1.5 for the C cohort, and 0.7 for the D cohort. Younger subjects should only participate in technique training for weightlifting exercises. From our results, the stated reference values seem to be average. However, the reference values for the younger subjects must be considered with respect to their training experience, age, and body weight.

Attention should also be given to the group comparisons in this study because the strength values of all cohorts were significantly higher ($p \leq 0.001$) in the STG compared with the CG. It can be assumed that an additional strength training program would promote strength in motor development and further demonstrate the high level of trainability in youth (4). Faigenbaum et al. (7) found that an 8-week strength training program promoted strength in motor development of younger children. Valez et al. (23) obtained a similar finding in a study of adolescents after a 12-week strength training program. Neural adaptations are primarily responsible for training-induced strength gains during pre-adolescence. Training-induced strength gains during and after puberty in male participants may be associated with changes in neural and hypertrophic adaptations.

The focus for improving power actions among youth in competitive sports should be on helping the youth achieve the necessary strength. To support this assumption, it must be noted that the relationship between maximum strength and power output is shown for both low and high loads (14). In many sports, low loads, such as kicking and throwing, and high loads, such as body weight during sprints and jumps, must be accelerated (19,24). Individuals must possess sufficient strength to overcome or accelerate body mass. McBride et al. (12) found a small-to-moderate relationship between absolute strength in the back squat and sprint performance (9). After calculating SREL, the relationship between the 2 parameters must be determined (2,12). Similar to this finding, another investigation showed that after calculating SREL, there was a high correlation between the clean and jerk exercises and the snatch and power predictors (2,10). Therefore, the focus for many sports should not only be on the absolute values but also on the SREL values.

In summary, the measured strength values show the high level of trainability of strength in youth and lead to reference strength values for young athletes. Long-term periodized strength training is recommended for young athletes. The

earlier that an individual begins the long-term periodized strength training, the easier and faster it is for that individual to reach the recommended reference values. In general, if a child is ready for participation in sport activities (generally 7 or 8 years of age), then he or she is ready for some type of resistance training (4). At this age, the first step should be creating a technical base for the weightlifting exercises. As training experience increases, the lifted weight can also increase, but the focus should continue to remain on maintaining the perfect technique. By age 11–12, players may have up to 4 years of training experience, and the lifted weight can be increased in small increments. By age 12–14, the lifted weight can be raised. This slow and continuous strength education has the aforementioned advantages and may potentially help individuals become professional athletes.

PRACTICAL APPLICATIONS

This investigation showed that young athletes can reach high values in 1RM and SREL in the parallel front and back squats with about 2 years of periodized strength training. The results of this investigation indicate SREL reference values in parallel squat for young athletes with training experience of 4–5 years to be a minimum of 2.0 for 16- to 19-year-olds, 1.5 for 13- to 15-year-olds, and 0.7 for 11- to 12-year-olds. Younger subjects should only participate in technique training for weightlifting exercises. Conversely, unsupervised, poorly performed strength training lacking periodization for young athletes is not recommended under any circumstances because of the potential for injury.

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